Ericsson PE Node ELEC-PE-06: Troubleshooting Hardware/Environmental Fault ALARM_4882

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_4882

Description

This Standard Operating Procedure (SOP) outlines the steps to identify and resolve hardware/environmental faults (ASIC memory parity errors, line card crashes, power supply redundancy loss) on Ericsson PE node ELEC-PE-06 at the Electronic City Hub. The fault is triggered by the ALARM_4882 code with a severity of Minor. The underlying protocol mechanics of this failure domain can be attributed to the complex interplay between the node's hardware components, specifically the Application-Specific Integrated Circuit (ASIC) memory parity errors, line card crashes, and power supply redundancy loss. The ASIC's memory parity error detection mechanism is designed to identify data corruption within the node's memory, while the line card crash is typically caused by a hardware or software fault. The loss of power supply redundancy may be due to a faulty or malfunctioning power supply unit. Understanding these hardware and environmental factors is crucial in diagnosing and resolving the issue. The purpose of this SOP is to provide a structured approach to troubleshooting and resolving this fault, thereby ensuring optimal network performance and availability.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Electronic City Hub for on-site support, if required.
- Verify the redundant link status and ensure it is operational before proceeding.
- Check the power supply units (PSUs) for any issues or faults.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to data corruption or hardware faults	High
Line card crashes caused by hardware or software faults	Medium
Power supply redundancy loss due to faulty or malfunctioning PSU	Low

Diagnostic Commands & Expected Outputs

```
Command: show card info, Expected Output: Card information (e.g., card type, status, error codes), Description: This command provides detailed information about the card, including its type, status, and any error codes that may be relevant to the fault. This helps to identify potential issues with the line card or ASIC.

Command: show power status, Expected Output: Power supply status (e.g., operational, faulty, malfunctioning), Description: This command checks the status of the power supply units and identifies any issues or faults that may be contributing to the power supply redundancy
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loss.

Command: show alarm details, Expected Output: Alarm details (e.g., alarm code, description, severity), Description: This command provides detailed information about the alarm, including the alarm code, description, and severity. This helps to confirm the presence and nature of the fault.
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Resolution Steps

Step 1: Step: Reboot the node, Command: reboot node, Description: Rebooting the node may resolve the issue by clearing any temporary faults or errors. Ensure that the node is properly shut down before rebooting.

Step 2: Step: Replace the faulty line card, Command: replace card , Description: If the line card crash is the root cause of the fault, replacing the faulty card with a spare or new one may resolve the issue.

Step 3: Step: Replace the faulty power supply unit, Command: replace PSU , Description: If the power supply redundancy loss is the root cause of the fault, replacing the faulty PSU with a spare or new one may resolve the issue.

Post-Check Verification

```
Command: show card info, Expected Output: Card information (e.g., card type, status, error codes) with no errors, Description: This command verifies that the card is operational and not reporting any errors.

Command: show power status, Expected Output: Power supply status (e.g., operational, no issues), Description: This command verifies that the power supply units are operational and not reporting any issues.

Command: show alarm details, Expected Output: Alarm details (e.g., no alarm, or alarm cleared), Description: This command verifies that the alarm has been cleared or is not present.
```

Roll-back Procedures

- Step: Revert changes made during resolution, Command: undo replace card , Description: If a line card was replaced, this command reverts the change by restoring the original card.

- Step: Revert changes made during resolution, Command: undo replace PSU , Description: If a power supply unit was replaced, this command reverts the change by restoring the original PSU.